

Maya, 32 · 10-year history of **severely painful periods, dyspareunia**, primary infertility · pelvic exam: tender nodules on uterosacral ligaments · laparoscopy: **blue-black implants on peritoneum, uterosacrals, and ovaries** · **endometriosis**

BIO 004 · WEEK 6 · CASE DEEP DIVE

The Female Reproductive System

A 32-year-old whose menstrual cycles are debilitating. Endometrial tissue where it shouldn't be. A diagnosis that took ten years.

PATIENT CASE

Maya is 32. Since her late teens she has had **severely painful periods**, missing work each cycle. The pain starts a few days before her period, peaks on day 1-2 of flow, and has gotten worse in the last few years. She has **deep dyspareunia** (pain with intercourse) and she and her partner have been trying to conceive without success for two years.

Pelvic exam reveals **tender nodules along the uterosacral ligaments**. Diagnostic laparoscopy shows **blue-black implants on the posterior cul-de-sac, both uterosacral ligaments, and the surface of both ovaries**; one ovary has a chocolate-colored cyst (endometrioma). Pathology confirms ectopic endometrial tissue and stroma. The diagnosis is **endometriosis**.

Endometriosis is the female reproductive system gone slightly wrong: tissue that belongs only in the uterine cavity grows outside it, responds to the same monthly hormones, and bleeds where it doesn't belong. Understanding it requires walking the female reproductive system from ovary to vulva.

GUIDING QUESTION

How is the female reproductive system organized, what happens at each location during the menstrual cycle, and why does endometrial tissue at the wrong site produce pain, infertility, and adhesions?

1 PLAN

Ovaries, tubes, uterus, vagina — and the cycle that links them

 Ovaries (2)

Almond-sized. Produce ova and hormones (estrogen, progesterone, small amounts of androgens).

 Uterine (fallopian) tubes

10 cm long. Receive the ovum, site of fertilization, transport to uterus.

 Uterus

Pear-shaped hollow muscular organ. Receives implanting embryo; cycles its lining monthly.

 Vagina

Muscular tube from cervix to outside. Birth canal, organ of intercourse, menstrual outflow.

 Vulva

External genitalia: labia majora, labia minora, clitoris, vestibule.

 Breasts

Modified sweat glands. Lactation under hormonal control.

Cortex full of follicles, medulla full of vessels

Each ovary is suspended by ligaments (suspensory, ovarian, mesovarium). The cortex contains **ovarian follicles** at every stage of development. A girl is born with all the oocytes she will ever have (~1 million); ~400 will ovulate in her reproductive life.

● Primordial follicle

Resting oocyte surrounded by a single layer of squamous granulosa cells.

● Primary → secondary follicle

Granulosa cells become cuboidal, multiply, and produce a fluid-filled cavity (antrum).

● Graafian (mature) follicle

Large fluid-filled vesicle. Ovulation: LH surge ruptures the follicle, releasing the secondary oocyte.

● Corpus luteum

Empty follicle transforms into a temporary endocrine gland producing progesterone (maintains endometrium if pregnancy).
Degenerates into corpus albicans if no pregnancy.

Where fertilization happens

Each tube is divided into **infundibulum** (funnel with fimbriae draping over the ovary), **ampulla** (longest segment, site of fertilization), **isthmus** (narrow segment near uterus), and **uterine (intramural) portion**. Lined by ciliated columnar epithelium plus peg cells (secretory). Cilia + smooth muscle peristalsis move ovum toward uterus.

A fertilized ovum that fails to migrate properly can implant in the tube → **ectopic pregnancy** (covered on the Pregnancy & Birth page).

Three layers and a cycle

Perimetrium

Outer serosa.

Myometrium

Thick smooth muscle. Contracts during labor and during menstruation.

Endometrium

Inner mucosa. Two layers: *functional layer* (sheds with menstruation, rebuilds) + *basal layer* (permanent, regenerates the functional layer).

Regions: **fundus** (above tube entries), **body** (main portion), **cervix** (narrow inferior part projecting into the vagina). Cervical canal lined by columnar epithelium; vaginal portion (ectocervix) lined by stratified squamous. The transition zone is where most cervical cancers arise (and where Pap smears sample).

Cycle: *menstrual phase* (functional layer sheds, days 1-5), *proliferative phase* (estrogen rebuilds endometrium, days 6-14), *secretory phase* (after ovulation; progesterone makes the lining receptive, days 15-28).

External and final passages

Vagina: muscular tube, ~8-10 cm, lined by non-keratinized stratified squamous epithelium. Rugae (folds) allow distension. Resident lactobacilli keep pH acidic (~4.5), inhibiting pathogens.

Vulva: *mons pubis* (fat pad over pubic bone), *labia majora* (outer folds), *labia minora* (inner folds), *clitoris* (erectile tissue, dense innervation), *vestibule* (space between labia minora, contains urethral and vaginal openings, Bartholin's glands).

Lobes, ducts, and a story for lactation

Each breast has 15-20 **lobes** radiating around the nipple. Each lobe drains via a **lactiferous duct** to the nipple. Lobes contain alveoli (in lactating women) and ducts (in non-lactating). Adipose tissue between lobes determines breast size.

Lymphatic drainage: mostly to **axillary nodes** (~75%), some to parasternal/internal mammary nodes. Why breast cancer is staged by axillary node involvement.



DRAW ZONE 1 -- Sketch the female reproductive system in sagittal section. Label ovaries, uterine tubes, uterus (fundus/body/cervix), vagina, vulva.

Show the path of the ovum from ovary → tube → uterus and where fertilization happens (ampulla).

DRAW ZONE 2

 -- Diagram the menstrual cycle: label menstrual, proliferative, secretory phases. Then mark where endometriosis implants commonly land (ovaries, uterosacrals, posterior cul-de-sac).

Show how implants bleed cyclically wherever they land, causing inflammation and adhesions.

Endometrium in the wrong neighborhood

Endometriosis is endometrial-LIKE tissue (glands and stroma) growing outside the uterine cavity. Each implant responds to monthly hormones, bleeds locally, and triggers inflammation.

"Painful periods that worsen over years"

Implants cycle with the same hormones as the uterine endometrium. Local bleeding causes inflammation and pain. Repeat cycles cause progressive scarring and worsening pain.

"Deep dyspareunia"

Implants on uterosacral ligaments and the posterior cul-de-sac are stretched during intercourse. Pain is deep, positional.

"Infertility"

Pelvic adhesions distort tubal anatomy, may obstruct fimbrial ova pickup, and may alter peritoneal environment. ~30-50% of women with endometriosis have infertility.

"Chocolate cyst (endometrioma)"

Implant on the ovary bleeds cyclically into a contained space. Old blood becomes chocolate-colored. Can grow large enough to require surgical excision.

"Why diagnosis takes so long"

Pain severity doesn't correlate with disease burden on imaging. Definitive diagnosis requires laparoscopy with biopsy. Symptoms overlap with primary dysmenorrhea, IBS, interstitial cystitis. Average delay is 7-10 years.

"Treatment"

Hormonal suppression of cycles (combined OCPs, progestins, GnRH agonists) to limit implant bleeding. Surgical ablation/excision of implants for severe disease or infertility. Definitive: hysterectomy with oophorectomy if family complete.

Ovary, tube, uterus, implant. Maya's case is the female reproductive system in disarray when one tissue grows where it doesn't belong.

